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EXP-5

TITLE: Secondary structure prediction by Chou-Fasman and GOR method

AIM: To predict the secondary structure of Mycobacterium tuberculosis Ftsz protein using the Chou-Fasman and GOR method.

INTRODUCTION: The prediction of a protein's secondary, tertiary and quaternary structure from its primary structure is known as structure prediction. Secondary structure prediction is a set of techniques in bioinformatics that aim to predict the secondary structures of proteins and nucleic acid sequences based only on knowledge of their primary structure. ... Secondary structure prediction can refer as Protein structure prediction. A prediction's success is determined by comparing it to the results of the DSSP algorithm applied the protein's crystal structure; for nucleic acids and it can also be determined from hydrogen bonding patterns.

There are different techniques with which the protein secondary structure can be predicted.

1. Statistical method
 - a) Chou-Fasman method

- b) GOR-IV method
- 2. Nearest neighbors
 - a) NNSSP
 - b) SSPAL
- 3. Neural network
 - a) PHD
 - b) Psi-Pred
 - c) J-Pred
- 4. SVM
- 5. HMM

The two methods of secondary structure prediction that we will deal with in the experiment is given below.

The **Chou–Fasman method** is an empirical technique for the prediction of tertiary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters were derived for the appearance of each amino acid in each secondary structure type, and these parameters are used to predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein. The method is at most about 50–60% accurate in identifying correct secondary structures, which is significantly less accurate than the modern machine learning–based techniques.

Chou-Fasman algorithm: The Chou-Fasman algorithm is simple in principle. The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure. These parameters are measures of a given amino acid's preference to be found in helix, sheet or coil. Using these conformational parameters, one finds nucleation sites within the sequence and extends them until a stretch of amino acids is encountered that is not disposed to occur in that type of structure or until a stretch is encountered that has a greater disposition for another type of structure. At that point, the structure is terminated. This process is repeated throughout the sequence until the entire sequence is predicted.

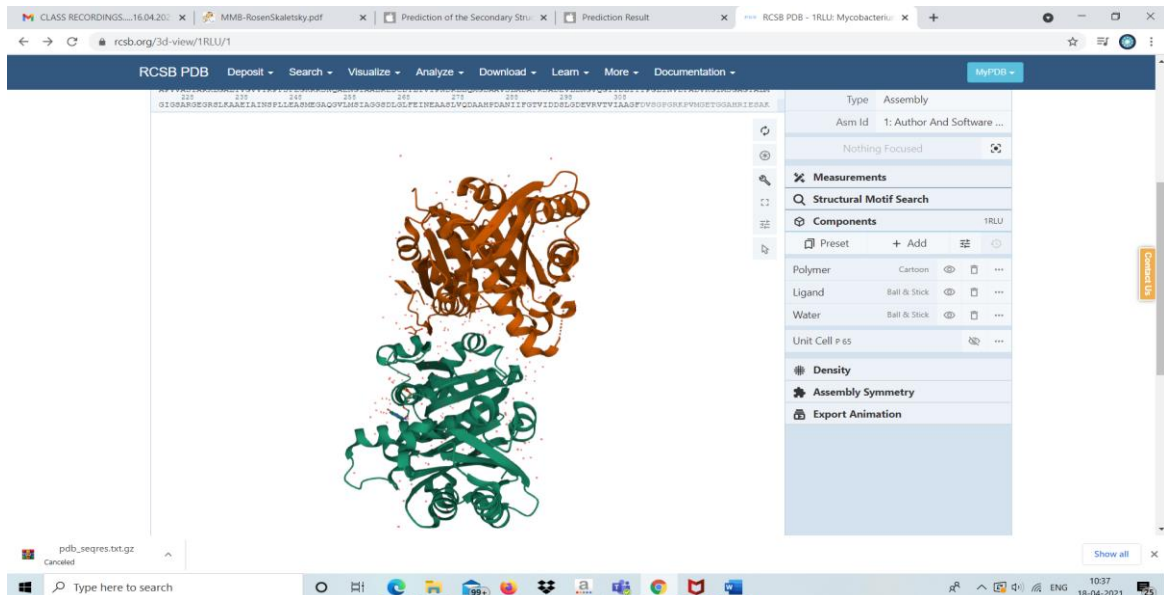
The **GOR method** (short for Garnier–Osguthorpe–Robson) is an information theory-based method for the prediction of secondary structures in proteins. It was developed in the late 1970s shortly after the simpler Chou–Fasman method. Like Chou–Fasman, the GOR method is based on probability parameters derived from empirical studies of known protein tertiary structures solved by X-ray crystallography. However, unlike Chou–Fasman, the GOR method takes into account not only the propensities of individual amino acids to form particular secondary structures, but also the conditional probability of the amino acid to form a secondary structure given that its immediate neighbors have already formed that structure. The method is therefore essentially Bayesian in its analysis.

GOR method algorithm: It uses the sliding window of 17 amino acid. The side amino acid sequence and alignment is determined to predict secondary structure of central sequence. It is good for helix than sheet because beta sheet has more inter sequence hydrogen bonding. There is 36.5% accuracy for beta sheet. In input we have to give any amino acid sequence and the output will tell about secondary structure.

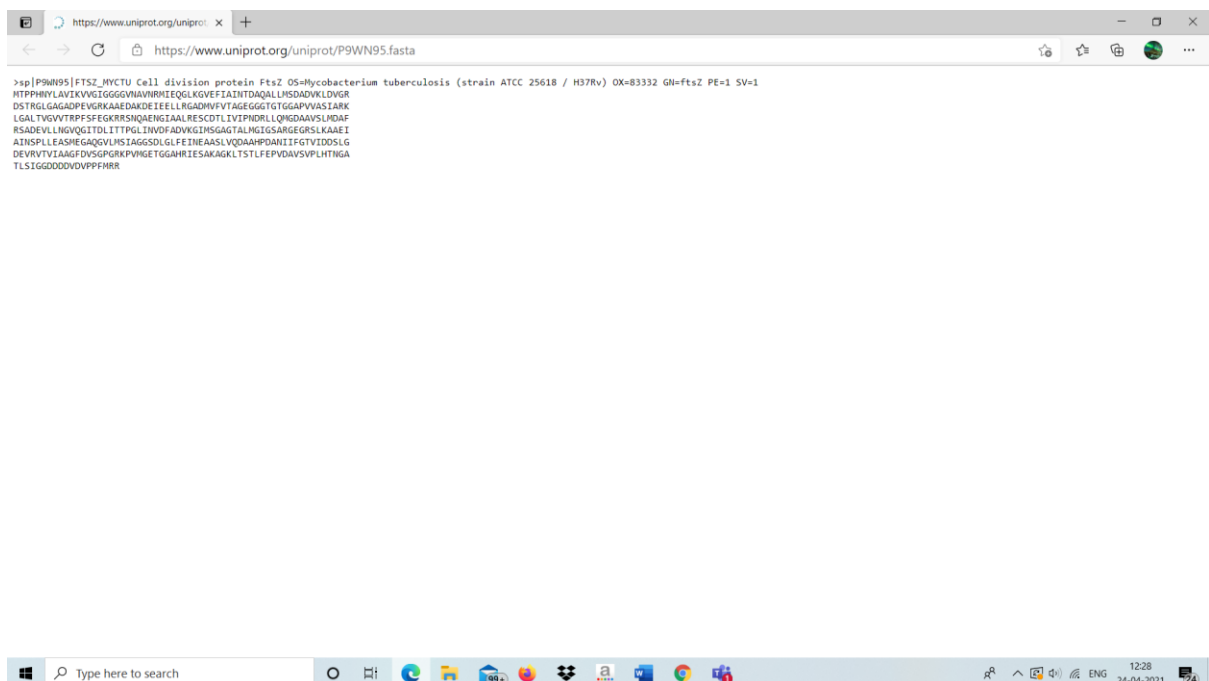
PROCEDURE:

- ❖ First we have to visit the website of secondary gene prediction by Chou Fasman, GOR and Neutral Network.
- ❖ Then we will go to uniprot and find the FASTA sequence of Ftsz protein of mycobacterium tuberculosis and copy that sequence and paste the sequence in the dialog box present in secondary gene prediction by Chou Fasman, GOR and neutral network website.
- ❖ We will now paste the FASTA sequence of the Fstz protein of Mycobacterium tuberculosis in the dialog box given in the secondary structure prediction dialog box. We will click both of the methods together (GOR and Chou Fasman method) and click on start prediction and output will appear which will show us the comparison between GOR method and Chou-Fasman method.
- ❖ Then we will go to <https://www.rcsb.org/3d-view/1RLU/1> website which contains Mycobacterium tuberculosis FtsZ in complex with GTP-gamma-S
- ❖ There we will download the PBD format and we will obtain a notepad file which contains some stretches of amino acid residues forming helix.
- ❖ The location of the helix which is given in the crystal structure was compared to the outputs that was found using the Chou Fasman and GOR method.

RESULT: The series of outputs are given below which are interpreted as the results.

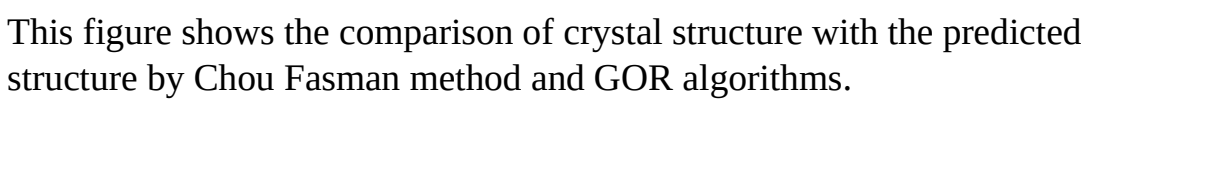


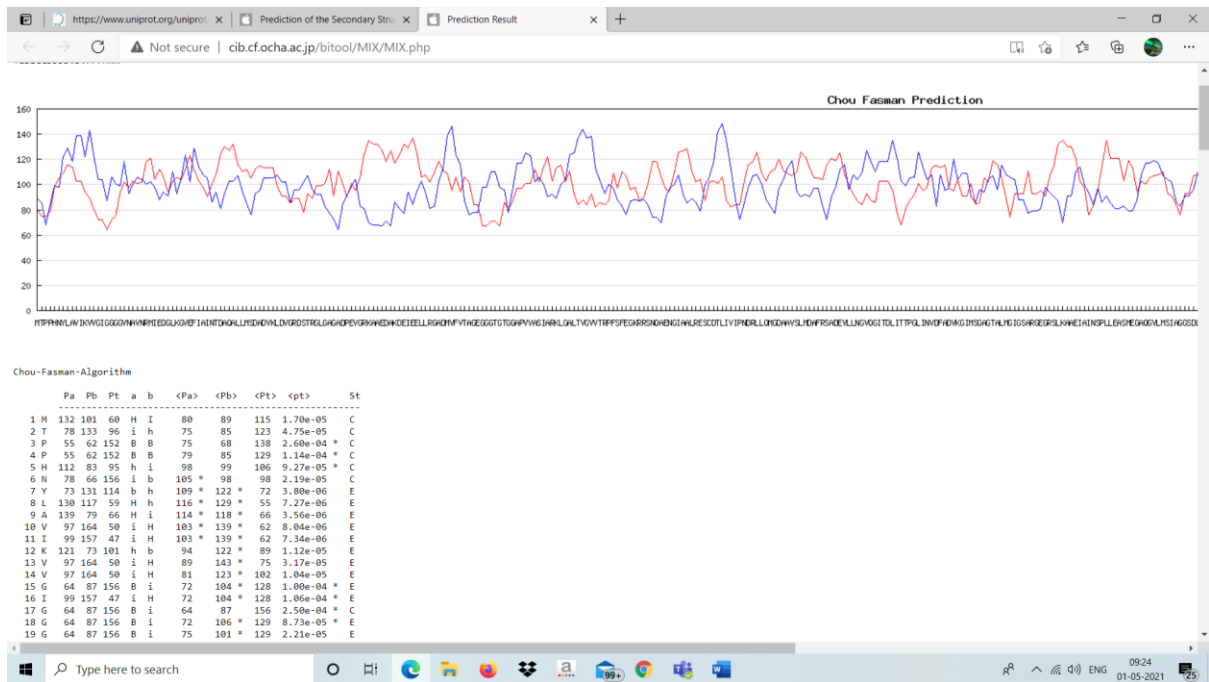
This figure shows the 3D structure from to <https://www.rcsb.org/3d-view/1RLU/1> website.



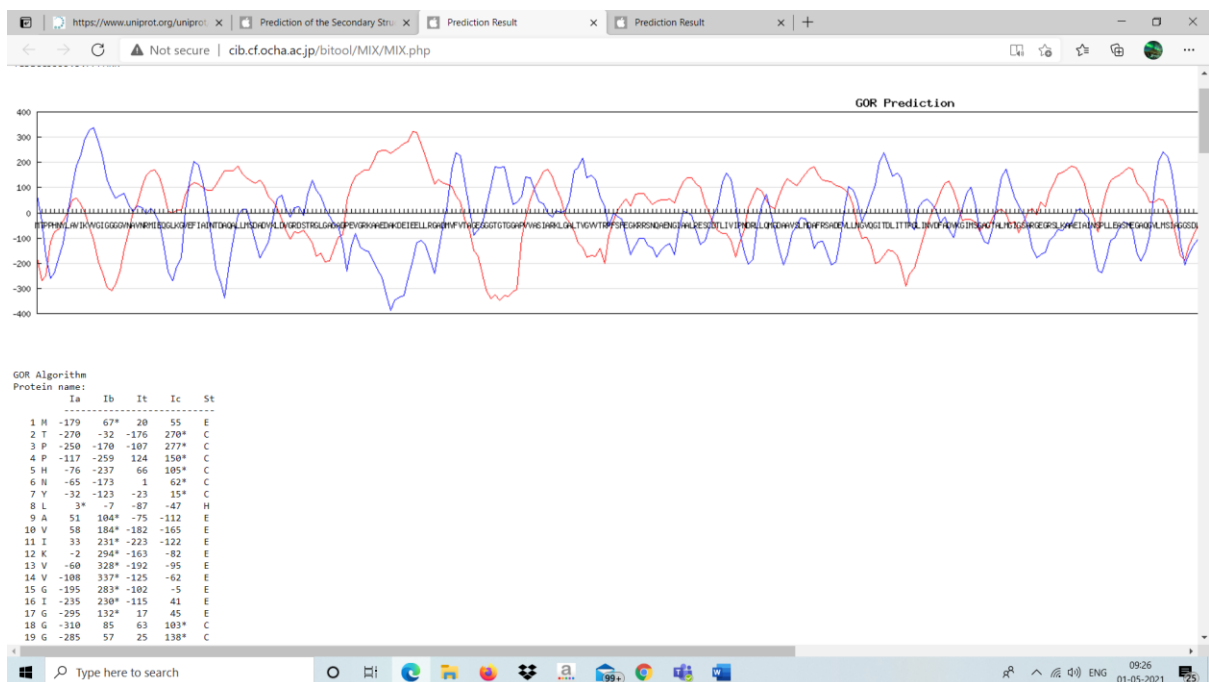
The screenshot displays the UniProt website showing the FASTA sequence for the protein FtsZ (P9WN95). The sequence is displayed in a text box, and the UniProt logo is visible in the top right corner.

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>sp|P9WN95|FTSZ_MYCTU_Cell division protein FtsZ 05-Mycobacterium tuberculosis (strain ATCC 25618 / H37Rv) OX=83332 GN=ftsZ PE=1 SV=1
HTPPHYLVAVKVGIGGGGVNAAWHTEGGLGVFFAINTDQALLHSDAPKLDVGR
DSTRGLGAGADPEVGRKAAEDAKDEIEELLRGDMVFVATGGGGTGTGGAPVVASIARK
LGALTGVGVTRPFSFEGKRRSNQAEIGIAALRESCDTLIVIPNDRLQLQGDAAVSLUDAF
RSADEVLLNGVQGITDLITTPGLTNVDFADVKIGMSGAGTALMGIGSARQEGRSLKAAEI
ATNSPLLEASHEGAQGVLSIAGGSDGLFETHEASLVQDAWPDANIIIFGTVIDDSLG
DEVRTVTVIAGFDVSGPRKPVINGETGGHRIESAKAGKLTSTLFEPVDVAVPLHTNGA
TLSIGGGDDDDVDVPPFHRH
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This figure tells us the secondary structure prediction by Chou-Fasman method



This figure tells us the secondary structure prediction by GOR method

CONCLUSION: From these pictures and observation we can conclude that the secondary structure of a protein can be predicted by GOR method and Chou-Fasman method.